Cylindrical cloak design in layered nanosphere dispersed liquid crystal metamaterial

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ABSTRACT

We discuss the concept of infrared cloaking using nanosphere dispersed liquid crystal (NDLC) matematerial in cylindrical geometry for TM polarization. The system consists of layers of NDLC with different values of ordinary refractive index and the same value of extraordinary refractive index of liquid crystal host. Finite element calculations (COMSOL Multiphysics), which include the Poynting vector calculations, show that scattering from the hidden object is limited in the presence of the layered cloak.

1. CLOAKING EFFECT – AN OUTLINE

The concept of metamaterials was proposed in 1968 by Veselago[1] who has shown that Maxwell equations predict the existence of propagating solutions corresponding to negative values of refractive index. However, this concept was not further explored because materials with such unusual properties can not be found in nature. The interest in the field of metamaterials started to grow steadily since the first experimental realization by Pendry[2] in 1998. The used metamaterial, composed of thin metal wires and split-ring resonators, offered no possibility of tuning. Since the first experimental realization of metamaterial behavior many different approaches to obtain the negative effective refractive index were invented. An incomplete survey of some of the recent result can be found in a review paper.[3]

Nanosphere dispersed liquid crystal (NDLC) metamaterial[4] offers a possibility of tuning of refractive index by changing the spatial distribution of the orientations of nematic liquid crystal (NLC) molecules which, in turn, causes the change of the permittivity of the host and influences the effective parameters of the metamaterial. We have applied Monte Carlo simulations to study the perspectives for tunability resulting from a joint action of electrostatic field, anchoring forces and temperature.[5–7]

Recently we have proposed a design of an invisibility cloak using NDLC metamaterial.[8–10] The aim of this paper is to discuss some features of the layer-like design of invisibility cloak proposed recently in Ref.[10]

2. NDLC METAMATERIAL

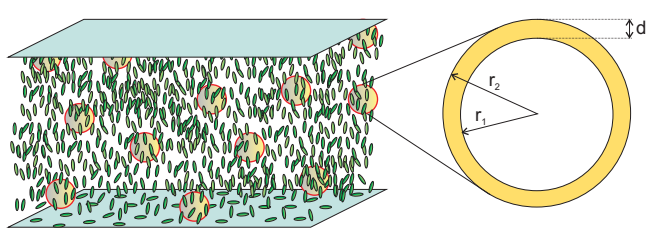

Figure 1. Model of the system: liquid crystal and dispersed nanospheres (left). Coated nanosphere (right).

In this Section we review briefly the concept of NDLC metamaterial following the presentation from Ref.[3]

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The studied NDLC metamaterial[4,11] is based on nematic liquid crystal in which the coated nonmagnetic nanospheres with core-shell structure are randomly distributed (Fig. 1). In this material the effective refractive index depends on the composition and dimensions of the coated spheres (Fig. 1), filling factor, and on the angle $\theta(\vec{r})$ which the long axis of the NLC molecule at point $\vec{r}$ makes with the direction of probing light. The core with radius r_1 is made from polaritonic material with relative permittivity

$$\epsilon_1(\omega) = \epsilon_1(\infty)\left(1 + \frac{\omega_L^2 - \omega_T^2}{\omega_T^2 - \omega^2 - i\omega\gamma_1}\right), \tag{1}$$

where $\epsilon_1(\infty)$ denotes the high-frequency dielectric constant, γ_1 – damping coefficient, ω_T and ω_L – transverse and longitudinal phonon frequencies, respectively. The relative permittivity of the semiconductor shell with outer radius $r_2 = r_1 + d$ is described by Drude model:

$$\epsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\gamma_2}, \tag{2}$$

where ω_p denotes plasma frequency and γ_2 – the damping coefficient.

Permittivity of the liquid crystal host for extraordinary polarized wave depends on angle θ:[11]

$$\epsilon_3(\theta) = \frac{\epsilon_\parallel \epsilon_\perp}{\epsilon_\parallel \cos^2(\theta) + \epsilon_\perp \sin^2(\theta)}, \tag{3}$$

where $\epsilon_\parallel$ and $\epsilon_\perp$ are the relative permittivities for light polarized respectively parallel and perpendicular to the director axis.

The effective permittivity ε_{eff} and permeability μ_{eff} of NDLC system were calculated using the method presented in papers[4,12] in the formalism of Mie scattering and Maxwell Garnet mixing rule:[13–15]

$$\varepsilon_{eff} = \varepsilon_3\left(\frac{k_3^3 + 4\pi i N a_1}{k_3^3 - 2\pi i N a_1}\right),\ \mu_{eff} = \frac{k_3^3 + 4\pi i N b_1}{k_3^3 - 2\pi i N b_1}, \tag{4}$$

where a_1 and b_1 are scattering coefficients,[13] $k_3 = \sqrt{\varepsilon_3}\,2\pi/\lambda$, λ denotes the free-space wavelength. N denotes number density of the spheres: $N = m/V_u$, where m stands for the number of spheres in a unit volume V_u. Formula (4) is valid in long-wavelength limit $\lambda/(r_1+d) >> 1$, for low values of filling factor f.[4]

3. INFRARED CLOAK IN NDLC

3.1 Cloak design in the absence of losses

In a recent paper[8] we have designed an infrared cylindrical TM cloak based on NDLC in the absence of losses. Consider a cylindrical NLC cell with inner and outer radii a and b, filled with NDLC metamaterial, with planar azimuthal alignment of NLC molecules in $x - y$ plane (Fig. 2 (a)). The incident light with transverse magnetic (TM) polarization ($\vec{E}$ in $x - y$ plane and $\vec{B}$ along the axis of the cylinder) propagates in $x - y$ plane, see Fig. 2 (b) and impinges as an extraordinary wave onto the NLC host.

The research done in Ref.[16] leads to the conclusion that for the cloak design it suffices to ensure proper values of the products $(\mu'_{eff})_z(\varepsilon'_{eff})_\Theta$ and $(\mu'_{eff})_z(\varepsilon'_{eff})_r$ (here Θ and r denote polar coordinates in $x - y$ plane, and primes - real parts):

$$(\mu'_{eff})_z(\varepsilon'_{eff})_r = \left(\frac{b}{b-a}\right)^2\left(\frac{r-a}{r}\right)^2, \tag{5}$$

$$(\mu'_{eff})_z(\varepsilon'_{eff})_\Theta = \left(\frac{b}{b-a}\right)^2. \tag{6}$$

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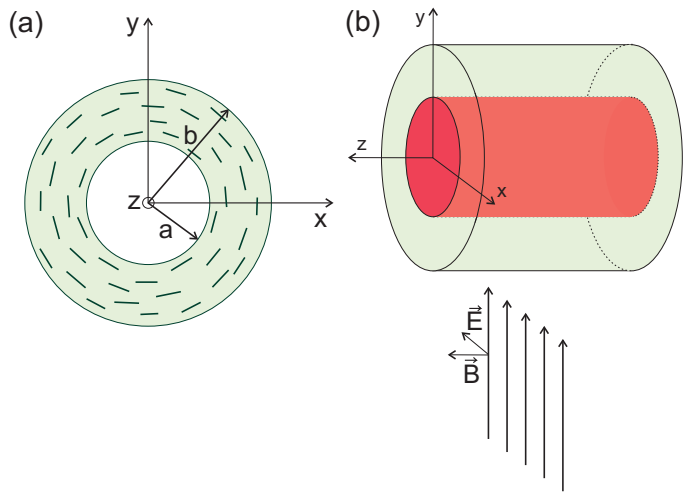

Figure 2. Planar cylindrical NLC cell (a) and geometry of the system (b).

The tensor components of $\varepsilon'_{eff}, \mu'_{eff}$ are calculated using the corresponding tensor components of ε_3. In cylindrical coordinates the tensor ε_3 corresponding to planar azimuthal alignment of NLC molecules in $x-y$ plane (Fig. 2 (a)) is diagonal with components $\varepsilon_r = \varepsilon_z = \varepsilon_\perp = n_o^2, \varepsilon_\Theta = \varepsilon_\parallel = n_e^2$. Real parts of radial and azimuthal components of effective parameters were calculated from Eq. (4): $(\varepsilon'_{eff})_r$ was calculated using $\varepsilon_3 = \varepsilon_r$ and $(\varepsilon'_{eff})_\Theta$ – using $\varepsilon_3 = \varepsilon_\Theta$.

The ultimate goal of the design was to find the variation in space of NLC host parameters n_o, n_e inside the cloak domain which guarantee that cloaking relations (5) – (6) hold. At the same time, the spatial dimensions of the cloak a, b and the operating wavelength λ_0 have to be fixed. The later was calculated from Eq. (5) applied at the inner boundary a, and taking into account Eq. (4): $\lambda_0 = 27.7\,\mu$m. Then, the effective permeability was independent on ε_3 in the interval of values relevant for this design: $(\mu'_{eff})_z \equiv \mu'_{eff} \simeq 1.012$. In this way the most challenging issue of the design was removed (similarly as in Ref.[17]). Correspondingly, the cloaking conditions (5) – (6) simplified noticeably: $(\varepsilon'_{eff})_\Theta$ should be constant inside the cloak and $(\varepsilon'_{eff})_r$ should vary with r according to (5). The former one requires that, in our geometry, $\varepsilon_\parallel = n_e^2$ should be constant. We have used $n_e^2 = 3$; then, Eq. (4) yields $(\varepsilon'_{eff})_\Theta \simeq 1.91$ for $\lambda_0 = 27.7\,\mu$m. The inner radius of the NLC shell ("invisibility area") was chosen to be $a = 20\,\mu$m. Then, formula (6) yields the outer radius of the shell $b = 71.2\,\mu$m. To complete the design we have ensured such a dependence of n_o on radial coordinate r that the product $(\mu'_{eff})_z(\varepsilon'_{eff})_r(r)$ is equal to the right-hand-side of Eq. (5). In Fig. 3 dependence of n_o on r is presented.

To evaluate the performance of this cloak design we have made field-mapping calculations using Finite Element package (COMSOL Multiphysics). The object hidden inside the cloak is an ideal metallic cylinder with radius a. The outgoing waves were absorbed at all the boundaries in the simulation domain. Fig. 3 shows the z-component of magnetic field distribution around the metallic cylinder surrounded by the NDLC cloak. Without the NDLC shell (Fig. 3 (a)) a shadow is cast behind the cylinder while in the presence of the cloak (Fig. 3 (b)) the wave fronts flow around the cloaked region with a rather small perturbation. The power reflection due to the impedance mismatch at the outer cloak boundary was estimated using the approach presented in Ref.[17] to be about 10%.

The quantitative characterization of the effect using the Poynting vector on a line (plane) behind the cloak confirms the effectiveness of the cloak design Fig. 4.

3.2 Layer-like design

In the experimental realization the variation of ordinary index $n_o(r)$ along r coordinate can be approximated by a stepwise distribution and implemented using cylindrical NLC layers separated by IR transparent walls, with constant values of refractive indices n_o and n_e in each layer. Fig. 5 shows an example of such a layered system. The cloak consists of 6 separated cylindrical cells with the same thickness. Parameters of liquid crystal

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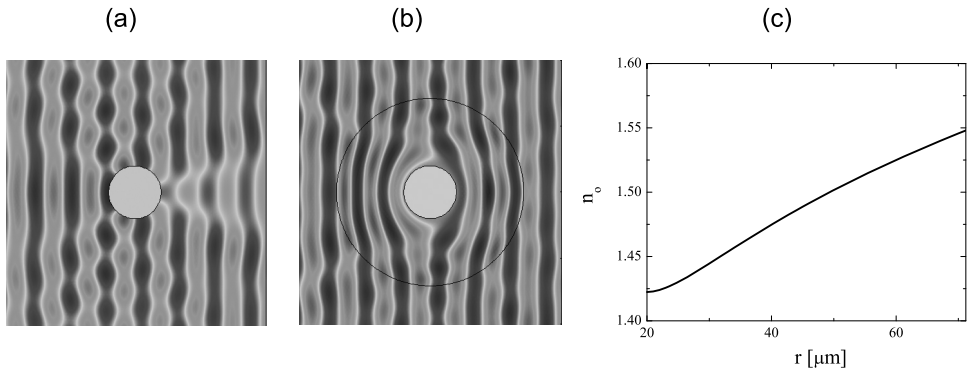

Figure 3. Finite Element calculations of the magnetic field mapping around the cloaked object with TM illumination: the object is surrounded by vacuum without the cloak (a); the object is inside the NDLC shell cloak (b); dependence of ordinary refractive index n_o on radial coordinate r (c).

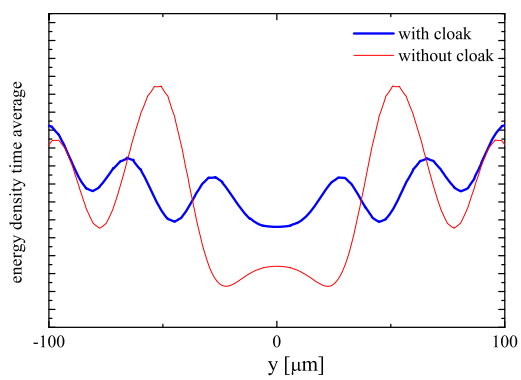

Figure 4. The quantitative characterization of the cloaking effect using the Poynting vector on a line (plane) behind the cloak.

and related effective parameters of NDLC are presented in Table 1. Size of the hidden object is $a = 20$ μm, size of the cloak $b = 71.2$ μm and the operating wavelength is $\lambda_0 = 27.7\mu$m. Fig. 6 shows the z-component of magnetic field distribution around the metallic cylinder surrounded by the NDLC shell, calculated using COMSOL software. In the absence of the NDLC shell (Fig. 6 (a)) a shadow is cast behind the cylinder. In Fig. 6 (b) the case of layer-like design ispresented. The quantitative comparison between those cases using the Poynting vector on line (plane) behind the object confirms the effectiveness of the cloak design. The optimization of cloaking effects requires numerical procedures[18,19] and will be presented elsewhere.

4. CONCLUSIONS

We have studied the cloak effect in the NDLC metamaterial composed of six layers with an exemplary distribution of ordinary and extraordinary indices of liquid crystal host. Finite Element simulations (COMSOL package) are

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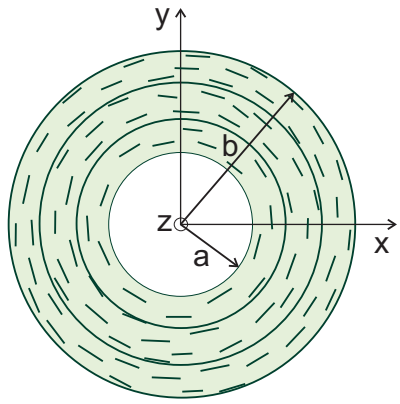

Figure 5. Model of the layered system. The desired distribution of the orientations of LC molecules results from anchoring forces on the cylindrical walls of layers.

layer	μ_z	$(\varepsilon'_{eff})_\Theta$	$(\varepsilon'_{eff})_r$	n_o	n_e
1	1.012	1.91	0.05	1.428	1.732
2	1.012	1.91	0.30	1.453	1.732
3	1.012	1.91	0.49	1.475	1.732
4	1.012	1.91	0.70	1.503	1.732
5	1.012	1.91	0.85	1.524	1.732
6	1.012	1.91	0.85	1.524	1.732

Table 1. Parameters used in the layer-like cloak design. The effective permittivity and permeability of NDLC were calculated for $r_1 = 800$nm, $d = 200$nm, $\lambda = \lambda_0 = 27.7\mu$m and $f = 0.06$.

used. We have shown, using the Poynting vector distribution behind the cloak, that the layered cloak design yields the desired result, which is however weaker than in the continuous theoretical distribution.

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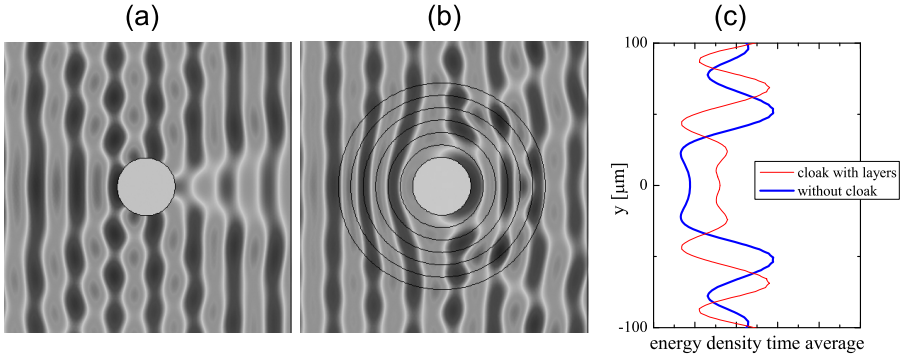

Figure 6. Finite Element simulations of the magnetic field mapping around the cloaked object with TM illumination: the object is surrounded by vacuum without the cloak (a); the object is inside layered NDLC shell cloak consisting of 6 layers (right). $\lambda_0 = 27.7\mu$m, $a = 20\mu$m, $b = 71.2\mu$m.

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